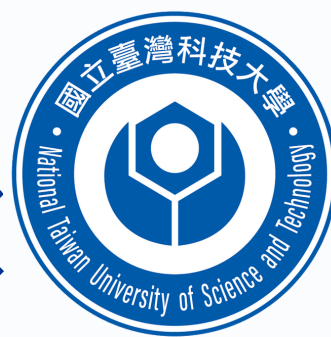




National Taiwan University of Science and Technology

AN EMISSION-AWARE EXTENSION OF THE CSPTW-TN-DO

**Original Paper: Crowd-Shipping Problem with Time Windows,
Transshipment Nodes, and Delivery Options**

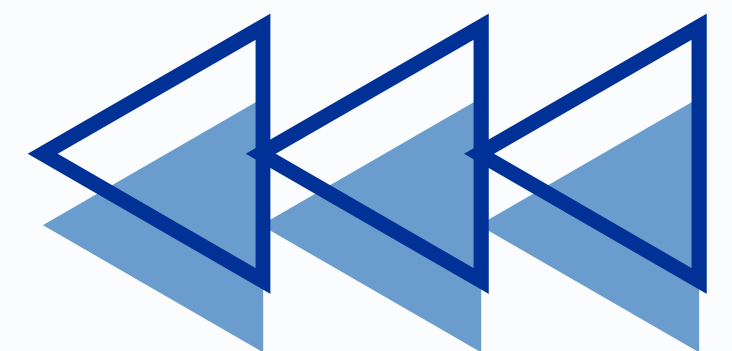
Optimization Models and Applications

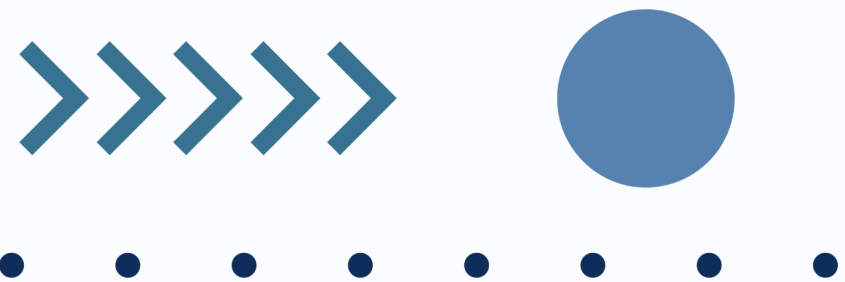
Professor: Chun-Chih Chiu

Group Members:

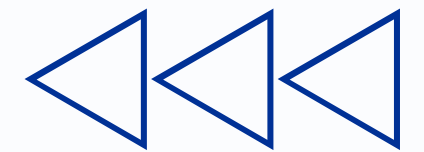
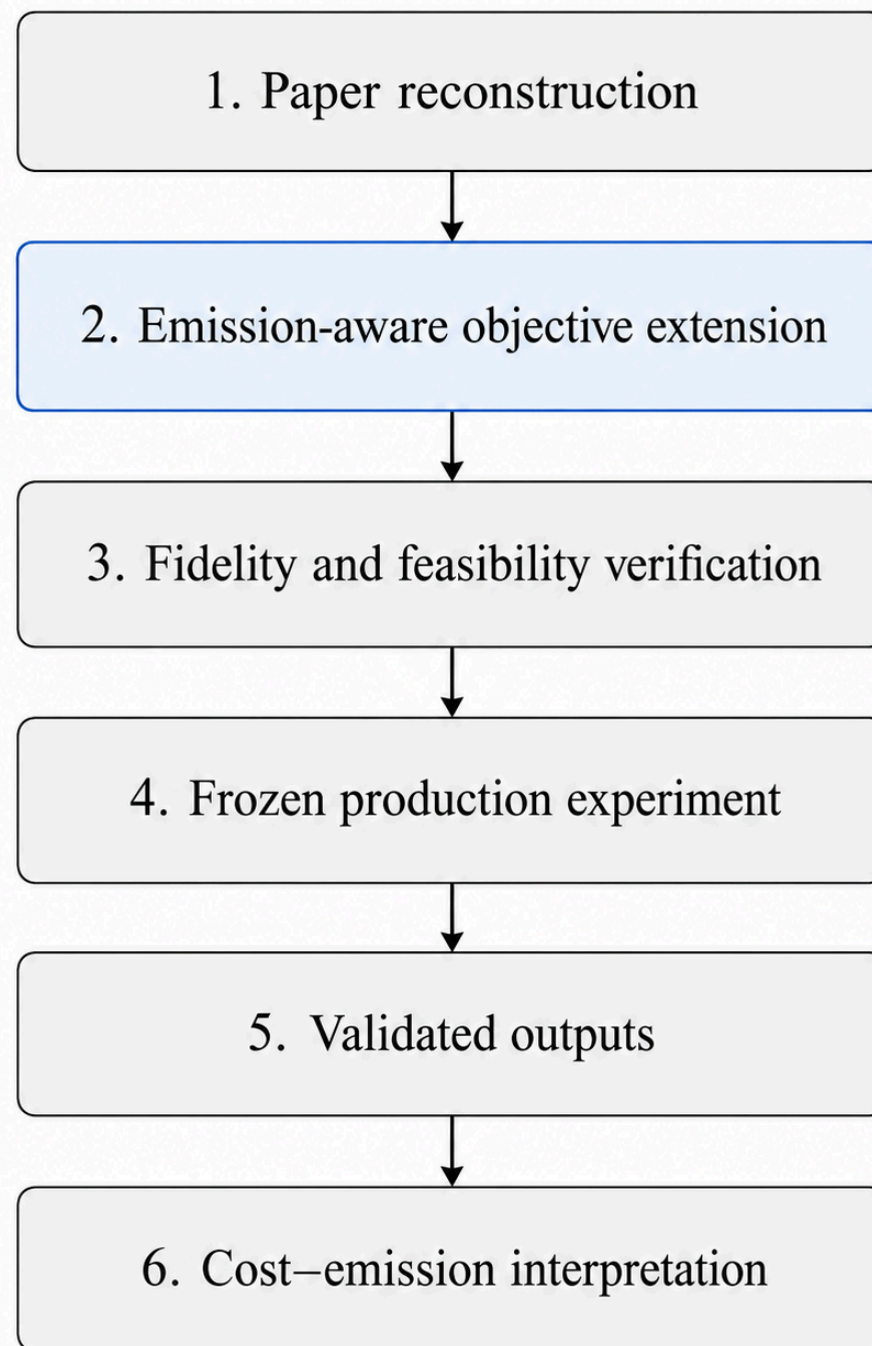
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STUDY WORKFLOW



ORIGINAL CSPTW-TN-DO SYSTEM

Main system elements:

- **Original problem:**
- **Dedicated vehicles**
- **Occasional drivers**
- **Transshipment nodes**
- **Alternative delivery points**
- **Time windows**
- **Three customer types**

Main challenge:

Routing decisions must satisfy delivery options, capacity, time windows, and TN synchronization

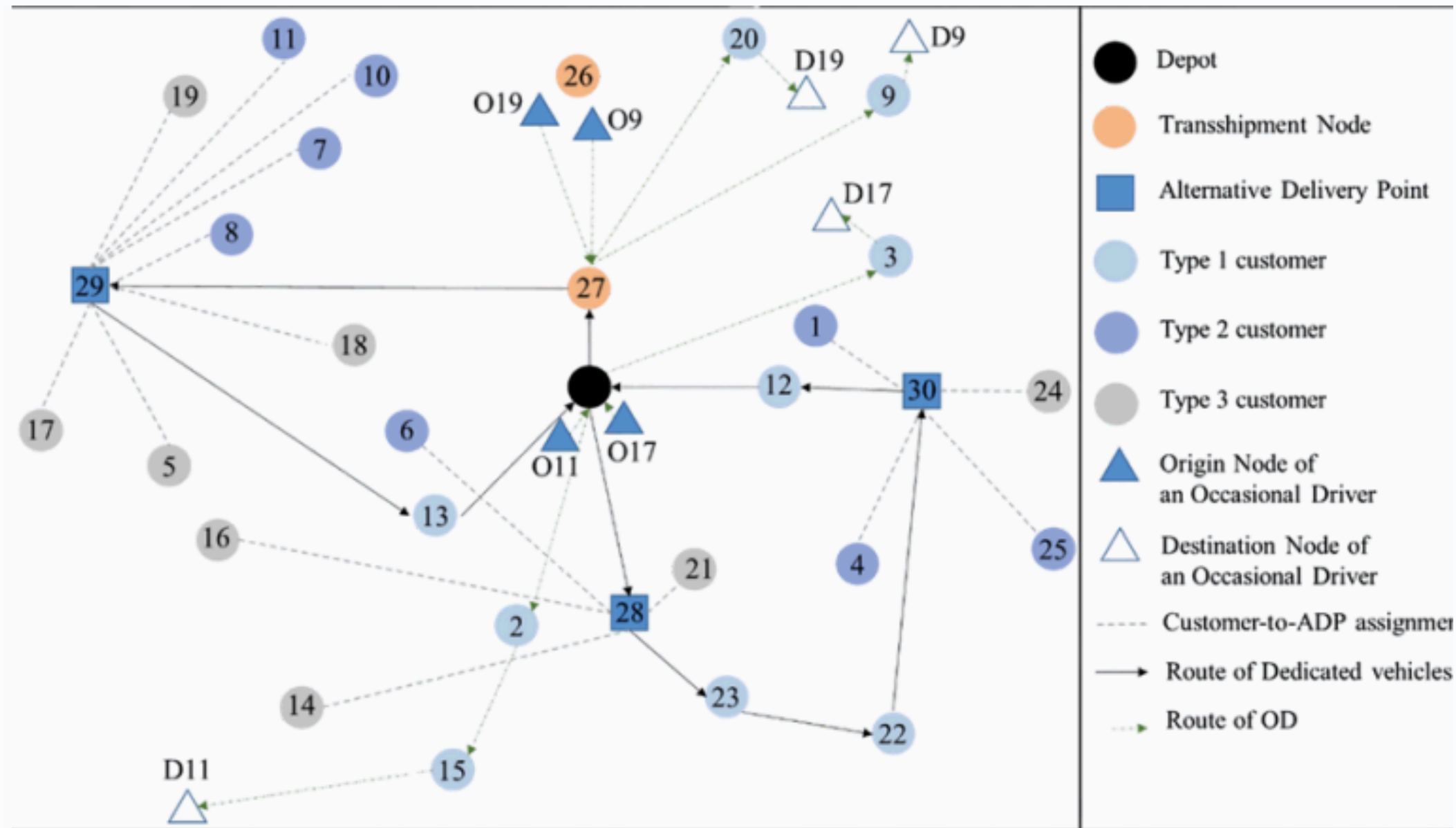


Fig 1. An illustrative example of a CSPTW-TN-DO solution.

FROM COST-ONLY OPTIMIZATION TO COST-EMISSION DECISION SUPPORT

Original paper:

- Objective: minimize total distribution cost
- Exact model + ALNS for CSPTW-TN-DO

Project extension:

- Keep the original delivery structure
- Add operational emissions
- Normalize cost and emissions
- Evaluate five λ settings

Important scope:

- Objective extension only
- No new ALNS is proposed

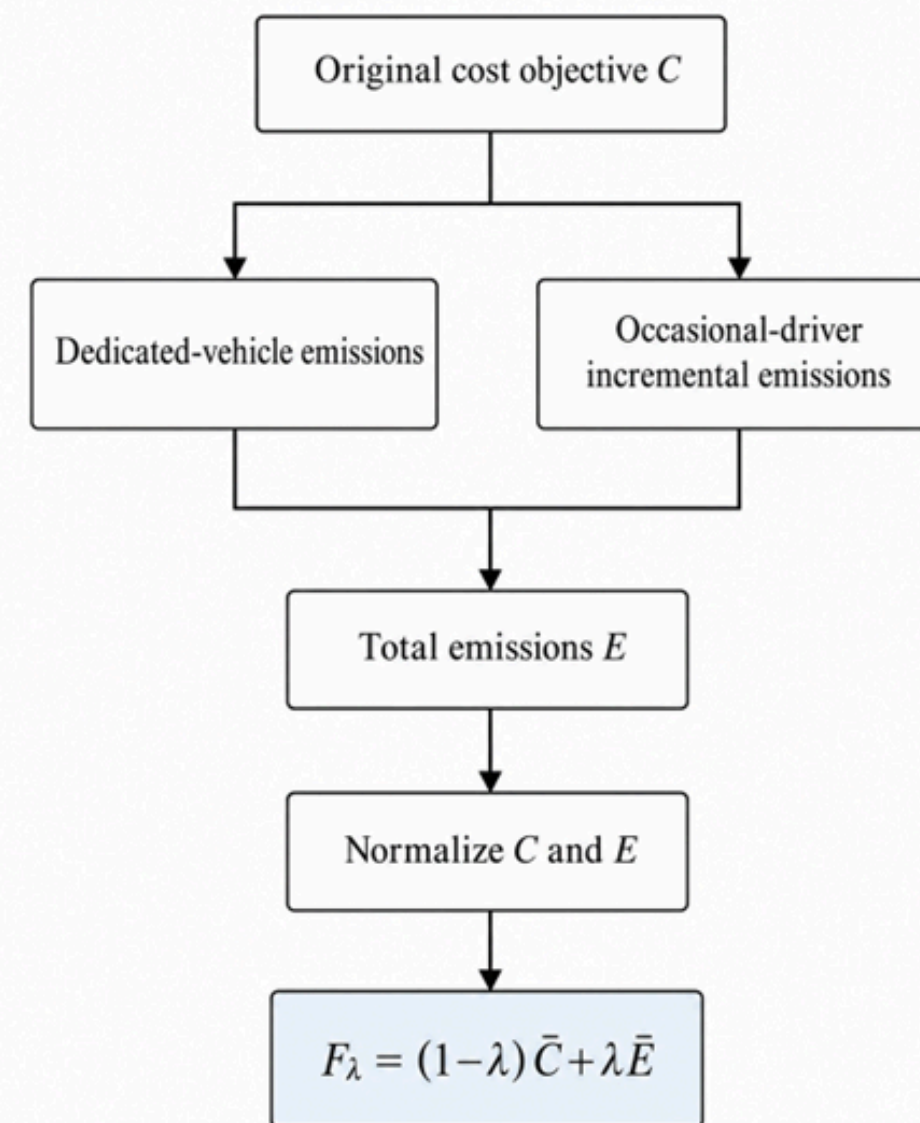


Figure 3. Extension from the original cost objective to the normalized cost-emission objective.

EMISSION-AWARE EXTENSION

Dedicated vehicle:

- Depot → delivery route → depot
- Full route counted
- $e_{DV} = 3.0$

Total operational emissions are calculated as:

$$E = E_{DV} + E_{OD} \quad (10)$$

or, in expanded form:

$$E = e_{DV} D_{DV} + \sum_{k \in CS} e_k^{OD} D_k^{extra} \quad (10)$$

Occasional driver

- Original trip: $O \rightarrow D$
- Used trip: $O \rightarrow$ delivery stop $\rightarrow D$
- Only extra detour counted
- $e_{OD} = 1.0$

The incremental-distance definition introduced in Section 2.3 is used:

$$D_k^{extra} = \max\{0, D_k^{used} - D_k^{direct}\} \quad (8)$$

where D_k^{used} is the route distance after accepting delivery tasks and D_k^{direct} is the original direct-trip distance.

NORMALIZED SCALAR OBJECTIVE

Why normalization?

- Cost and emissions have different units
- Raw values cannot be directly added
- One scale may dominate the objective

Weighted objective:

$F\lambda = (1 - \lambda) \text{ normalized cost} + \lambda \text{ normalized emissions}$

Lambda	Interpretation
0	Cost-oriented
0.25	More cost-focused
0.50	Balanced after normalization
0.75	More emission-focused
1.00	Emission-oriented

Normalized cost is:

$$\bar{c}(S) = \frac{[C(S) - C^{ideal}]}{[C^{nadir} - C^{ideal}]} \quad (15)$$

Normalized emissions are:

The normalized objectives are combined as:

$$F_{\lambda}(S) = (1 - \lambda)\bar{c}(S) + \lambda\bar{e}(S) \quad (17)$$

subject to:

$$0 \leq \lambda \leq 1 \quad (18)$$

A lower value of $F_{\lambda}(S)$ is preferred.

Lambda controls the priority between normalized cost and normalized emissions.

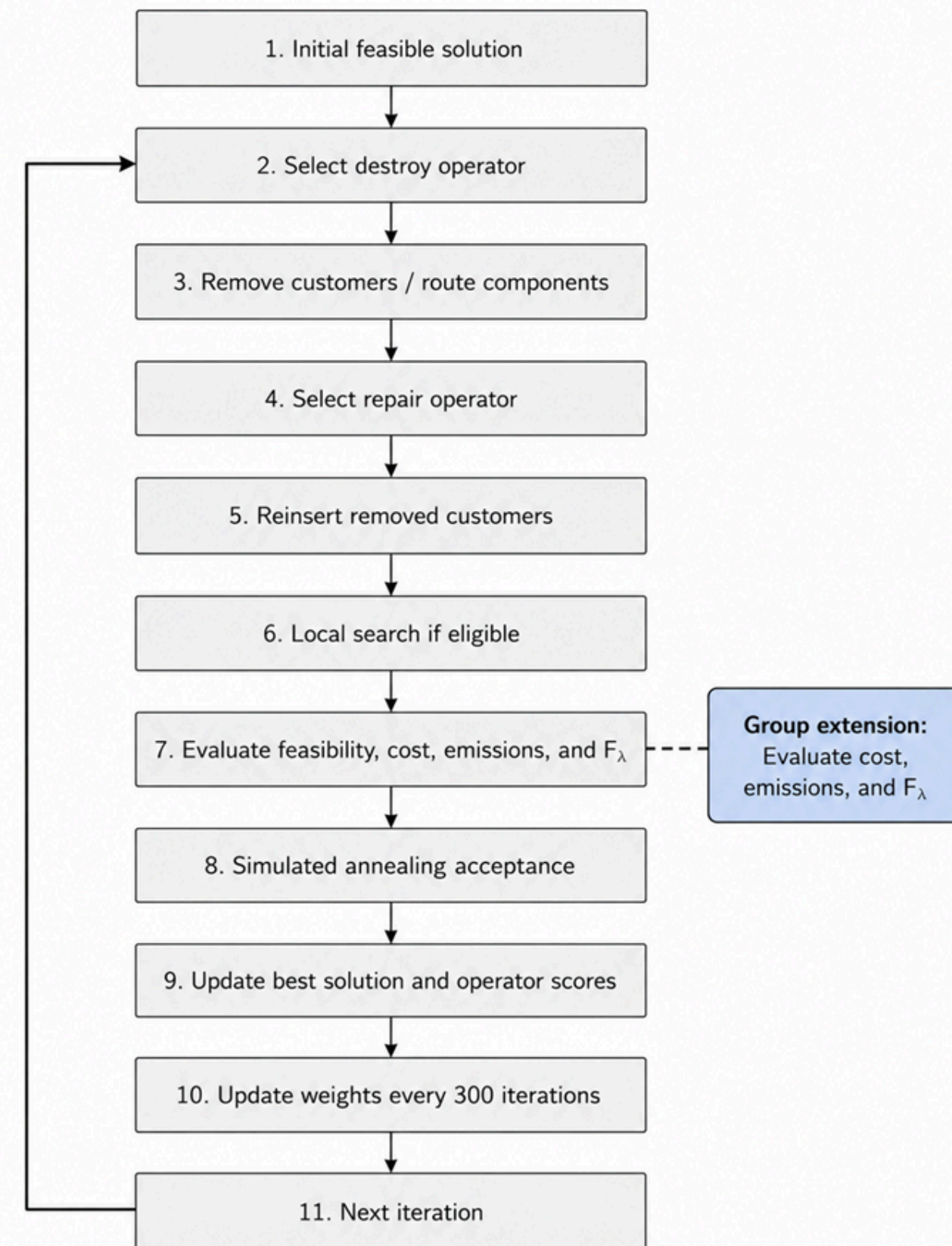
ALNS FRAMEWORK

Paper-based ALNS retained:

- Initial solution
- Destroy and repair
- Local search
- Simulated annealing acceptance
- Adaptive operator weights

Project-specific change:

Each feasible solution is evaluated by cost, emissions, and F_λ .



WHY WE TRUST THE OUTPUTS

Gate	What was checked	Evidence used	Status
Structural fidelity	Operator inventory, parameter values, and solution representation	Controlled component tests and configuration review	Pass
Behavioral fidelity	Customer types, delivery options, capacities, time windows, and TN synchronization	Shared final validator	Pass
Numerical consistency	Cost, emissions, normalization, and recomputation	Independent evaluator	Pass
Reproducibility	Lambda value, random seed, iteration count, configuration, and output files	Run manifests and deterministic seeds	Pass
Production readiness	Complete outputs, no duplicate runs, and final validator pass	Consolidated result checks	Pass

Key takeaway: Results were interpreted only after the algorithm structure, feasibility rules, objective calculations, and production outputs had passed the defined reliability gates.

EXPERIMENTAL DESIGN

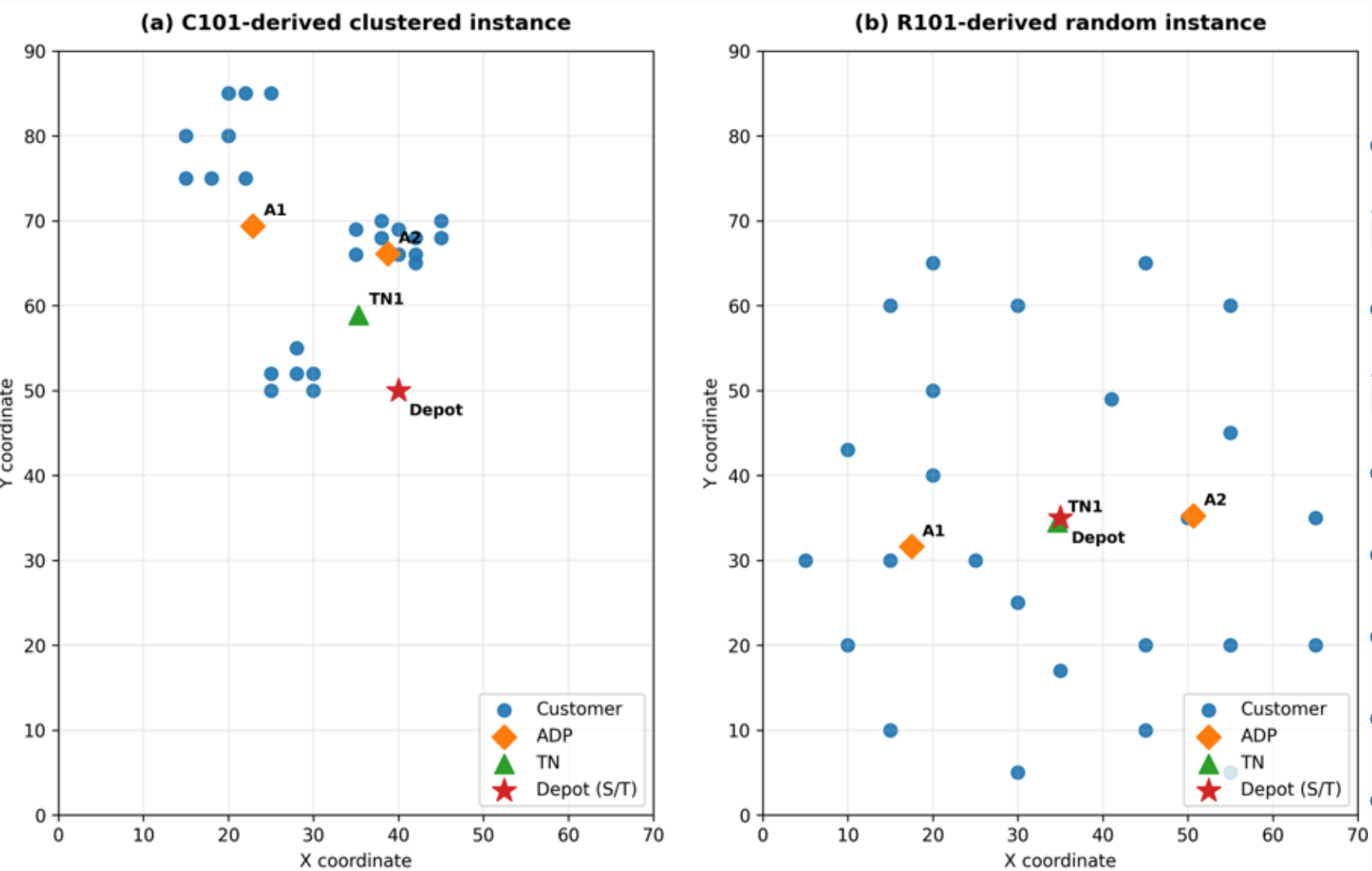


Figure 5. Spatial layouts of the C101-derived and R101-derived CSPTW-TN-DO instances.

Final ALNS benchmark:

- C101-derived clustered instance
- R101-derived random instance
- 25 customers each
- 5 λ values
- 12 matched seeds per λ
- 301 iterations per run
- 120 ALNS runs total
- Same commit, hardware, and seed set

Setting	C101-derived	R101-derived
Solomon source	C101	R101
Spatial structure	Clustered	Random
Customers	25	25
Lambda values	0, 0.25, 0.5, 0.75, 1	0, 0.25, 0.5, 0.75, 1
Seeds	2029-2040	2029-2040
Seeds per lambda	12	12
Iterations per run	301	301
Planned runs	60	60
Code revision	1079caf	1079caf
Hardware	MacBook Air M4	MacBook Air M4

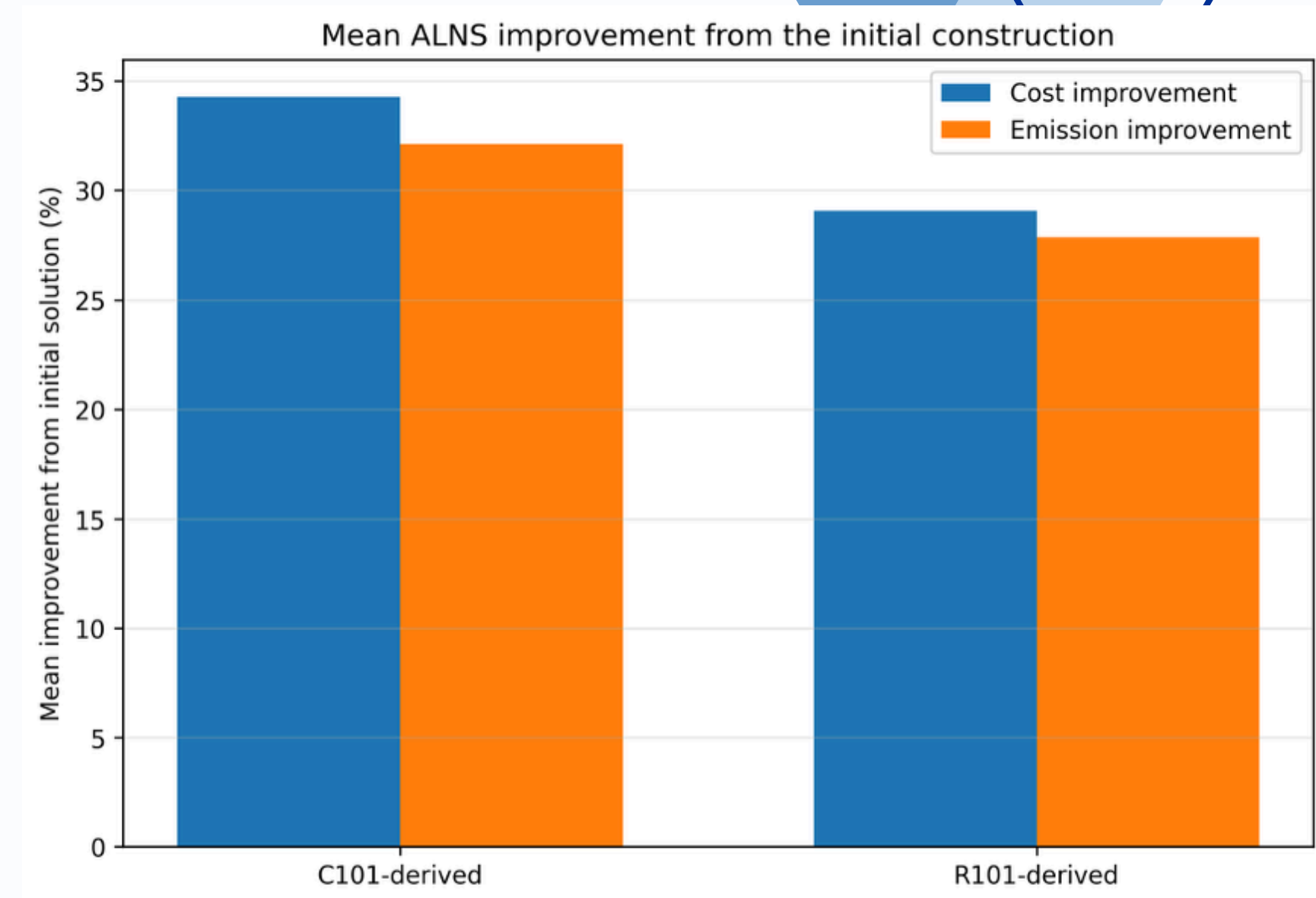
RESULT GATE: COMPLETE AND VALIDATED DATASET

Check	CI01-derived	RI01-derived
Expected runs	60	60
Completed runs	60	60
Unique lambda–seed combinations	60	60
Duplicate combinations	0	0
Validation pass rate	100%	100%
Iterations per run	301	301
Total runtime	Approx. 2.56 hours	Approx. 2.46 hours

Key takeaway: All 120 planned ALNS runs were completed, unique, and feasible for interpretation.

ALNS IMPROVEMENT FROM THE INITIAL CONSTRUCTION

Instance	Mean cost improvement range	Mean emission improvement range
C101-derived	33.67%–34.69%	31.38%–32.43%
R101-derived	28.2%–29.4%	27.3%–28.2%



Main evidence

- **ALNS** improved the initial feasible solution under every tested lambda.
- **C101-derived** achieved approximately 34% cost improvement and 32% emission improvement.
- **R101-derived** achieved approximately 29% cost improvement and 28% emission improvement.
- The larger **C101** improvement may reflect stronger consolidation opportunities in the clustered structure.

Key takeaway: The reconstructed ALNS produced substantial improvement before the effects of different lambda settings were compared.

RESULTS 1: C101 RESULTS

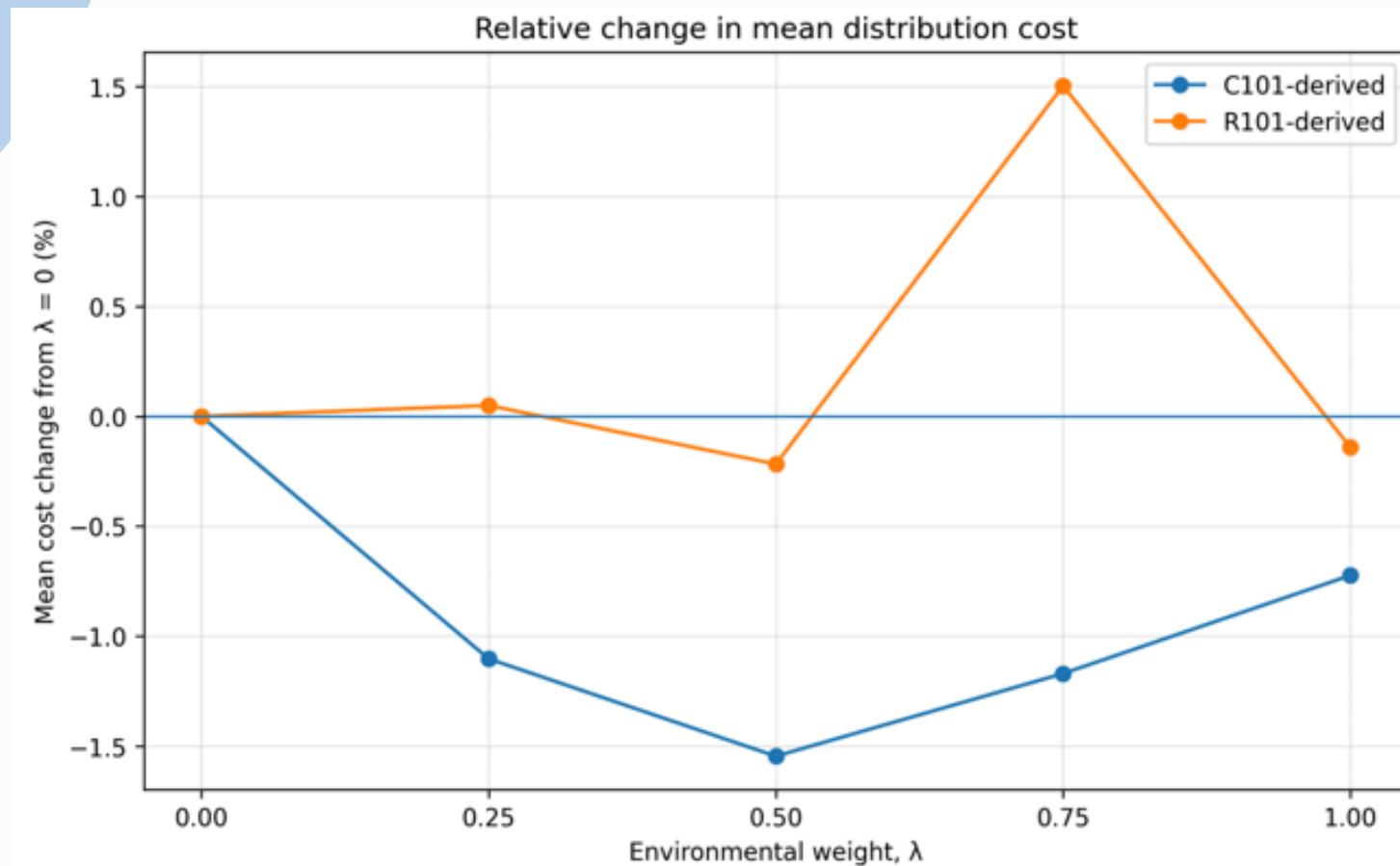


Figure 7. Relative change in mean distribution cost from the ($\lambda = 0$), baseline.

C101-derived clustered instance:

- All lambda values ≥ 0 reduced both mean cost and mean emissions compared with $\lambda = 0$.
- The lowest mean cost was obtained at $\lambda = 0.50$.
- The lowest mean emissions were obtained at $\lambda = 0.75$.
- Mean cost reduction at $\lambda = 0.50$ was about 1.55%.
- Mean emission reduction at $\lambda = 0.75$ was about 1.53%.
- The response was more stable than R101-derived.

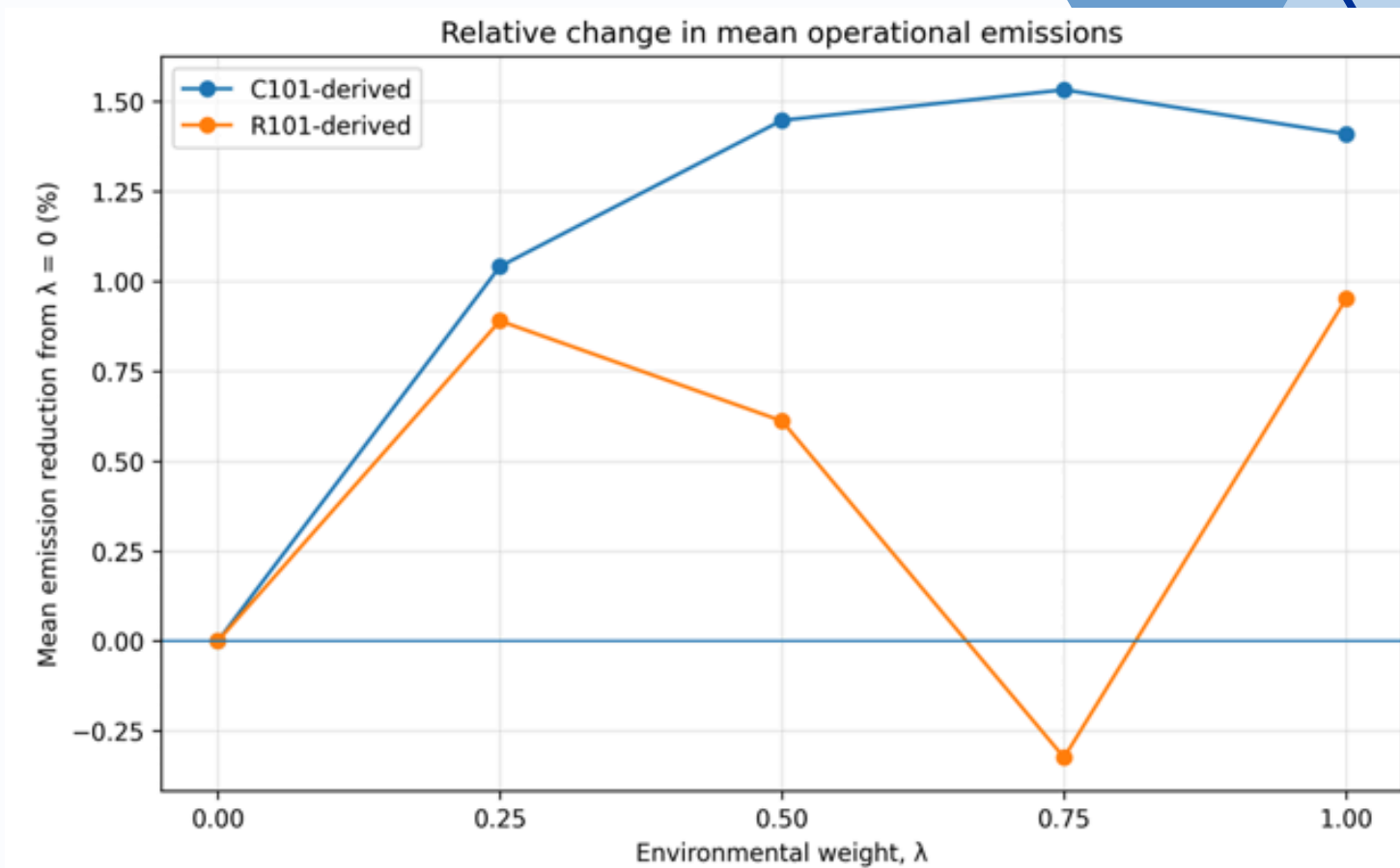


Figure 8. Relative change in mean operational emissions from the ($\lambda = 0$), baseline.

Interpretation:

- For the clustered instance, intermediate lambda settings helped ALNS find solutions that improved both cost and emissions.

Key message:

- Emission awareness can support both cost and emission improvement in a clustered delivery structure.

RESULTS 2: R101 RESULTS

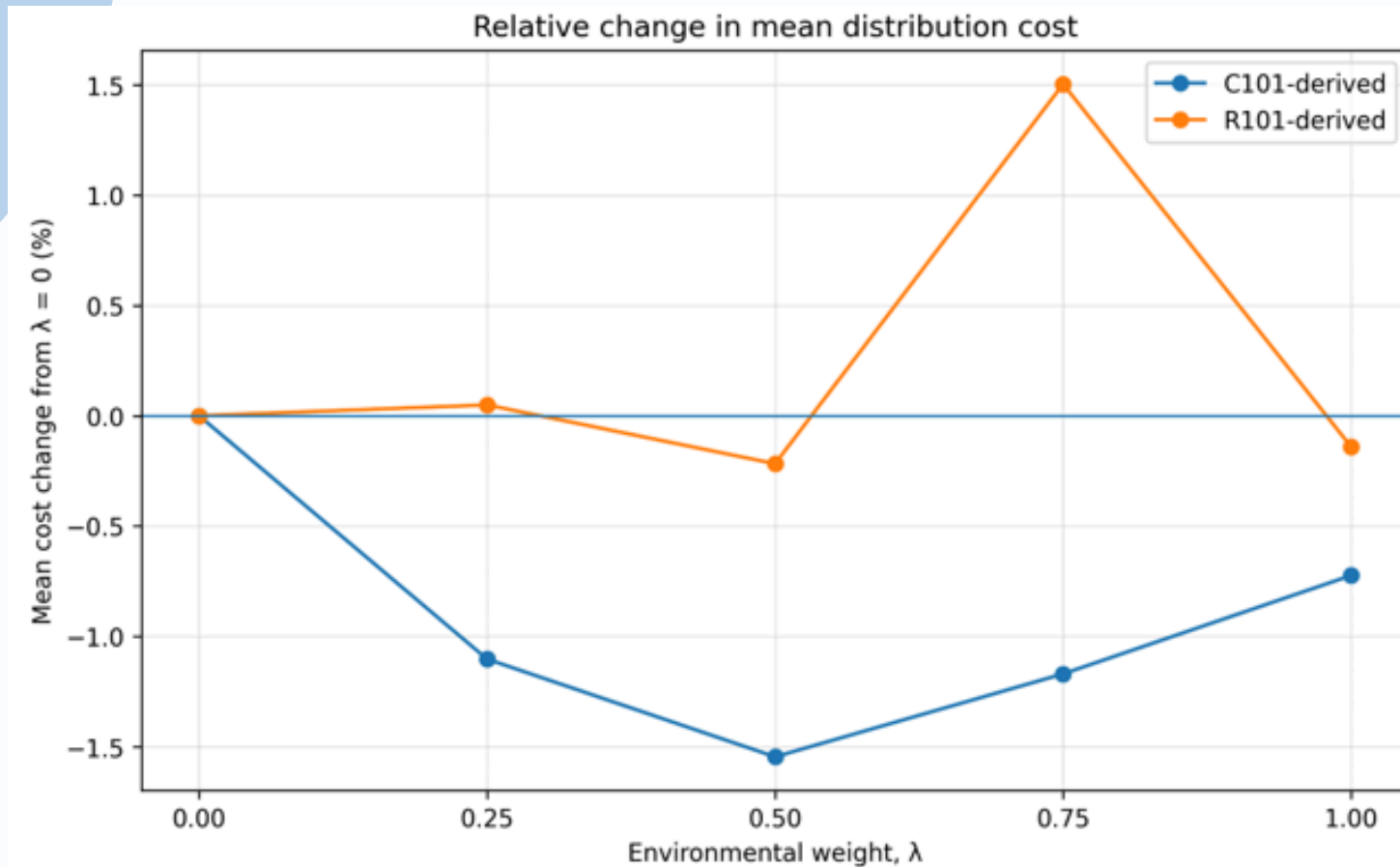


Figure 7. Relative change in mean distribution cost from the ($\lambda = 0$), baseline.

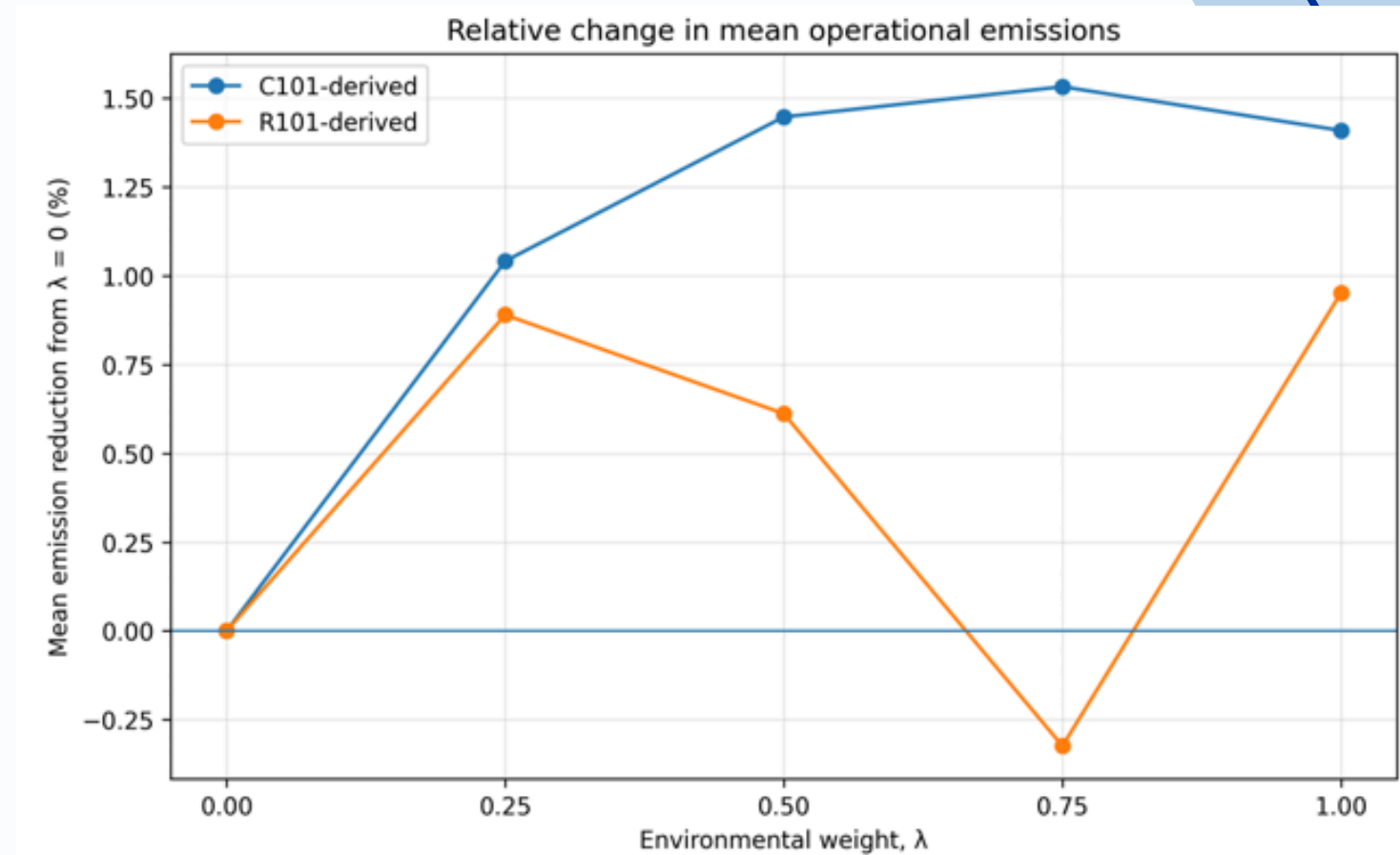


Figure 8. Relative change in mean operational emissions from the ($\lambda = 0$), baseline.

R101-derived instance:

- The response was weaker and non-monotonic.
- $\lambda = 1.00$ produced the lowest mean emissions.
- $\lambda = 0.50$ slightly reduced both mean cost and emissions.
- $\lambda = 0.75$ increased both mean cost and mean emissions compared with $\lambda = 0$.
- The results did not form a clean monotonic trade-off.

Interpretation:

- Increasing the emission weight does not always produce better emission performance under a short stochastic search budget.

Key message:

- The effect of lambda is less stable in the random spatial structure.

OBSERVED COST-EMISSION OUTCOMES

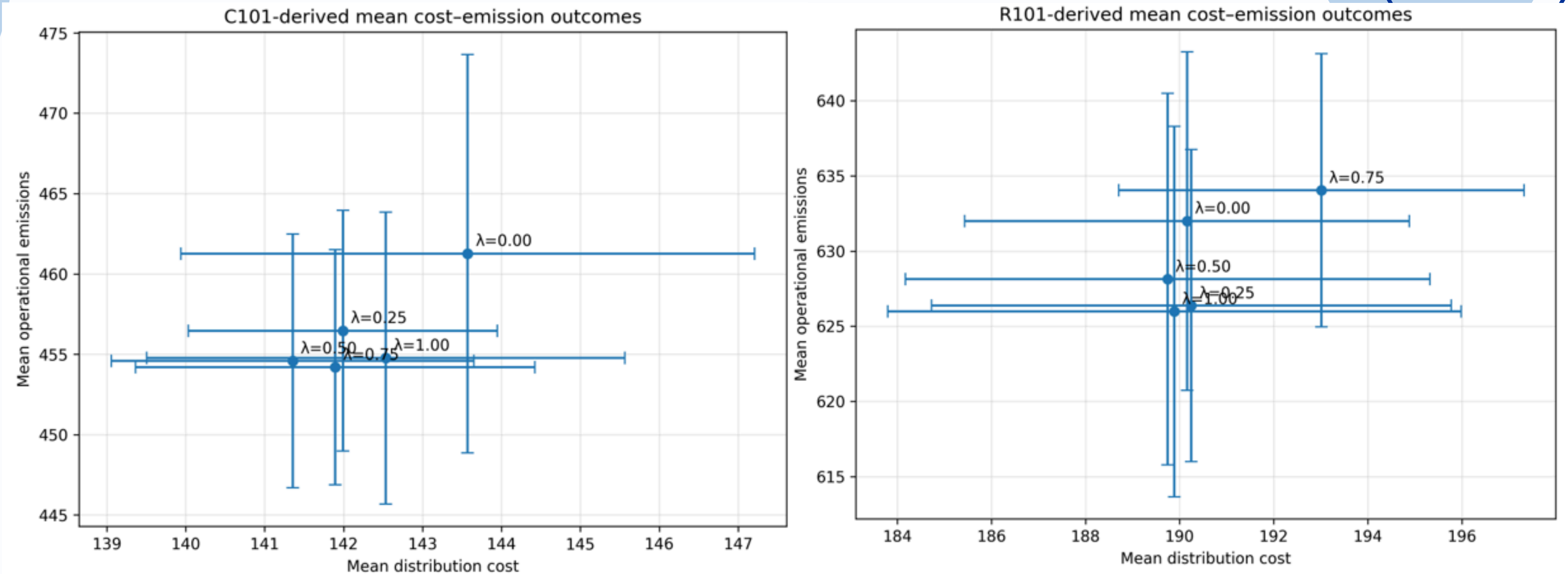


Figure 9. Mean cost–emission outcomes under the five lambda settings.

Key interpretation:

- C101 points moved more consistently toward lower cost and lower emissions.
- R101 points were more dispersed.
- These are weighted-sum outcomes, not a complete Pareto frontier.

WHAT THE EXACT SOLVER SUPPORTS

Exact solver role:

- Generated the C10I empirical anchors.
- Provided small-instance formulation references.
- Returned feasible weighted-sum incumbents.
- Weighted-sum runs did not certify zero-gap optimality.

Anchor run	Cost	Emissions	Runtime (s)	Solver status	Relative MIP gap (%)
Cost-oriented anchor	138.620394	450.764046	257.10–262.63	OPTIMAL*	4.62
Emission-oriented anchor	140.106516	445.860718	300.01	FEASIBLE	15.45–15.46

λ	Cost	Emissions	F_λ	Runtime (s)	Solver status	Relative MIP gap (%)
0	142.131	450.336	2.362	300.06	FEASIBLE	5.17
0.25	148.869	474.919	6.654	300.01	FEASIBLE	2.202
0.50	140.712	448.889	1.013	300.01	FEASIBLE	1.379
0.75	140.624	447.413	5.74	300.01	FEASIBLE	1.431
1.00	146.793	464.323	3.765	300.01	FEASIBLE	8.91

The exact solver supports anchor generation and formulation-level evidence, but the time-limited incumbents do not establish a complete cost–emission frontier.

CONCLUSION

1. The original **CSPTW-TN-DO** structure was retained.
2. Operational emissions were added for **DV** routes and **OD** detours.
3. Cost and emissions were normalized before weighting.
4. All **I20 ALNS** runs were completed and validated.
5. **CI0I** and **RI0I** responded differently to λ .
6. Cost and emissions can be partly aligned, but the effect depends on spatial structure and search performance.

Optimization can support greener last-mile decisions by evaluating distribution cost and operational emissions together, while recognizing that the best balance depends on the delivery network structure.

REPRESENTATIVE C101-DERIVED VALIDATED ROUTE

Configuration

- Instance: C101-derived
- Spatial structure: Clustered
- ($\lambda = 0.50$)
- Seed: 2034
- Iterations: 301
- Final validator: Pass

What the figure shows

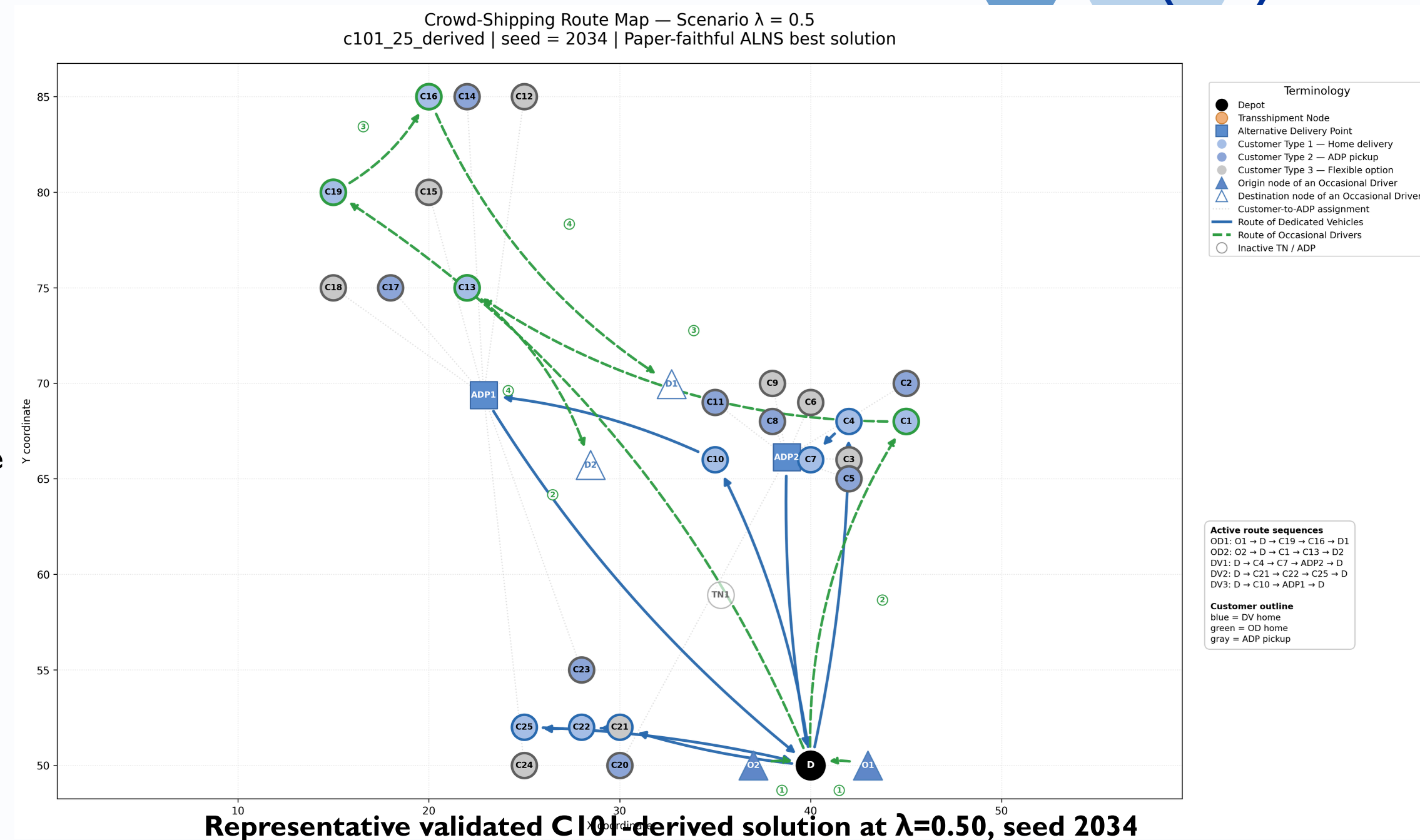
- Dedicated-vehicle and occasional-driver routing decisions
- Customer, ADP, TN, and depot locations
- A complete feasible solution under the balanced objective setting
- Visible opportunities for serving nearby locations through clustered routing and consolidation

Important interpretation

- The clustered structure may support shorter connections and stronger consolidation opportunities through nearby customers, ADPs, and TNs.

Evidence boundary

- This route is an illustrative validated solution. The reported C101 conclusions are based on all twelve seeds for each lambda, not on this single route



Key takeaway: The production output represents a complete feasible routing plan, not only an objective value.

REPRESENTATIVE R101-DERIVED VALIDATED ROUTE

Configuration

- Instance: R101-derived
- Spatial structure: Random
- ($\lambda = 0.50$)
- Seed: 2034
- Iterations: 301
- Final validator: Pass

What the figure shows

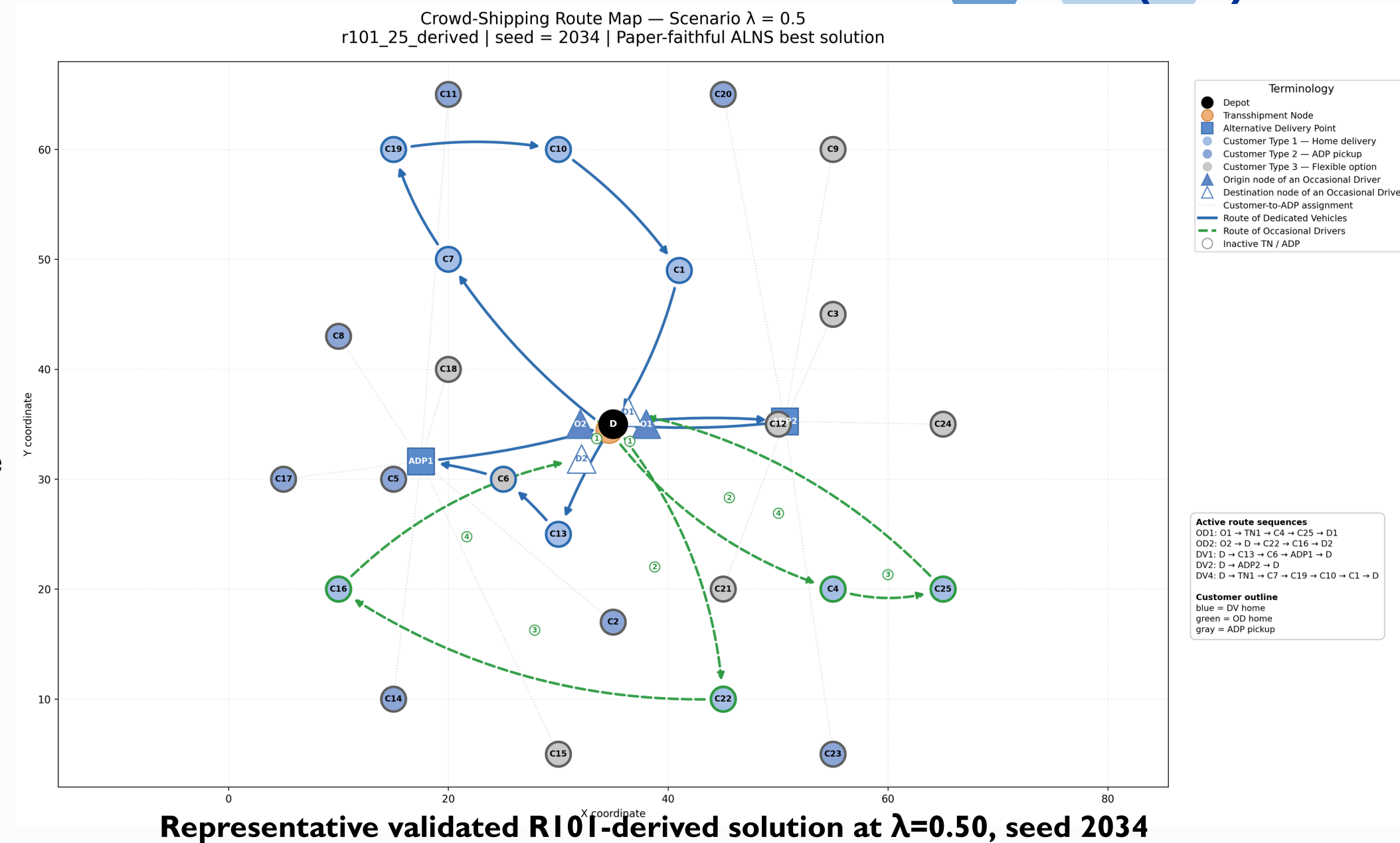
- Dedicated-vehicle and occasional-driver routing decisions
- Customer, ADP, TN, and depot locations
- A complete feasible solution under the same balanced objective setting
- More dispersed customer connections than in the clustered C101-derived case

Important interpretation

- The random spatial structure may create longer and more varied route connections, which can make the search response less stable across lambda settings.

Evidence boundary

- This route is an illustrative validated solution. The reported R101 conclusions are based on all twelve seeds for each lambda, not on this single route.



Key takeaway: The dispersed structure creates a different routing pattern, but the statistical conclusions come from the full multi-seed experiment.

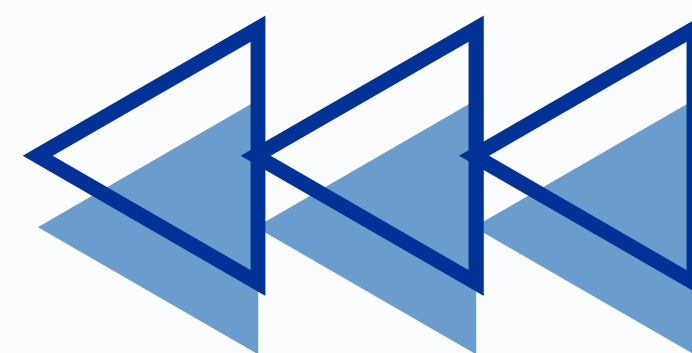
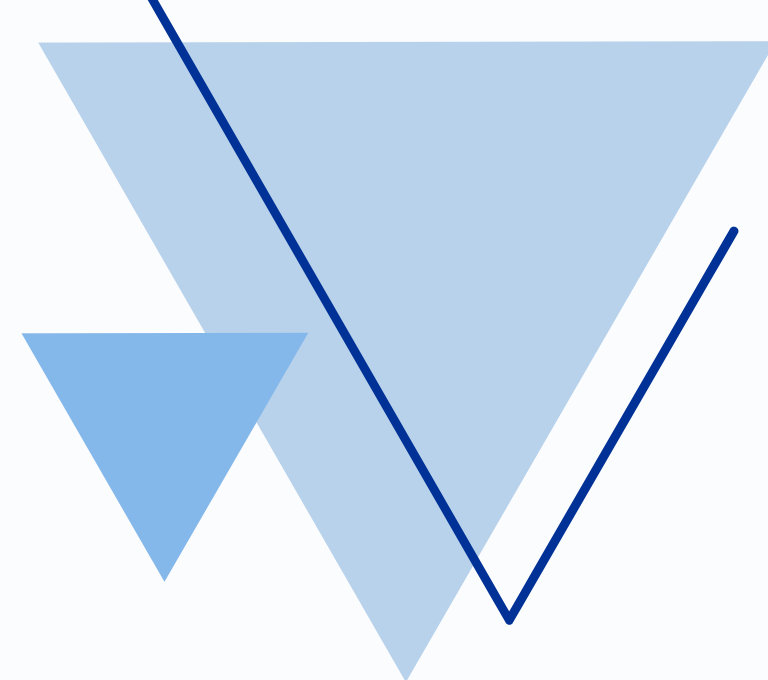
FUTURE WORK

- Increase the **ALNS** iteration budget to allow several adaptive-weight update segments.
- Extend the benchmark to larger **C**, **R**, and **RC** instances with multiple cases from each spatial family.
- Improve the emission model by including vehicle type, load, traffic, speed, and electric-vehicle characteristics.
- Apply Pareto-based or epsilon-constraint methods to examine the cost–emission frontier more directly.
- Develop one shared canonical evaluator for the exact solver, **ALNS**, and final validator.
- Test the framework using real operational data for customers, **ADPs**, **TNs**, occasional-driver trips, and emission factors.

Future work should strengthen computational depth, environmental realism, and exact–heuristic comparability.



Q&A





THANK YOU

